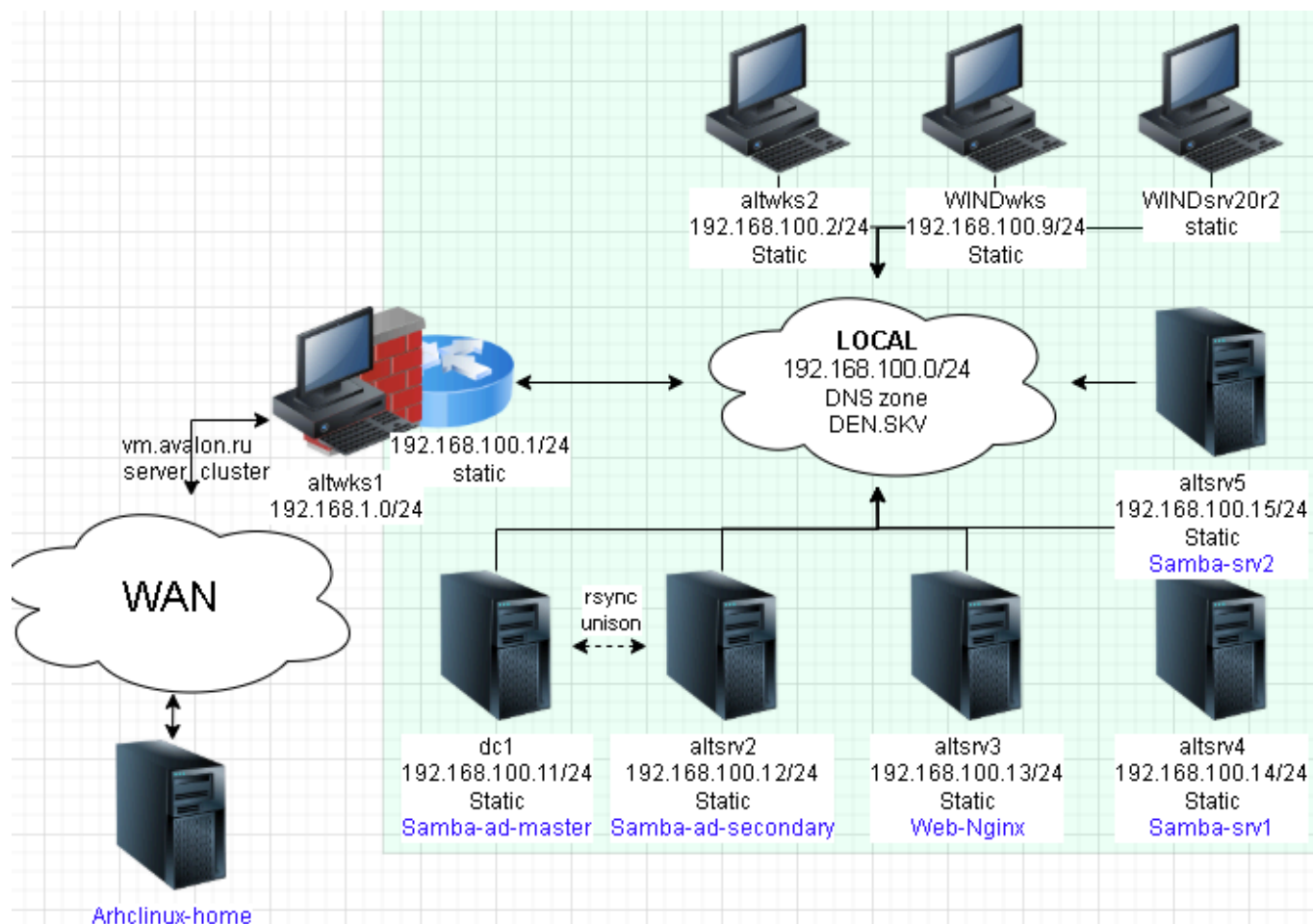


Лабораторная работа 7 «Настройки групповых политик домена»



Памятка входа

```
# Регистрация сгенерированного ssh агентом
eval $(ssh-agent) \
&& ssh-add \
~/.ssh/id_alt-domain_2026_host_ed25519

# Хост altwks1
> ~/.ssh/known_hosts \
&& ssh -t -o StrictHostKeyChecking=accept-new \
sysadmin@172.16.100.2 \
"su -"

# Хост dc1
ssh -t \
-i ~/.ssh/id_alt-domain_2026_host_ed25519 \
-J sysadmin@172.16.100.2 \
-o StrictHostKeyChecking=accept-new \
sysadmin@192.168.100.11 \
"su -"
```

```
# Xocr dc2
ssh -t \
-i ~/.ssh/id_alt-domain_2026_host_ed25519 \
-J sysadmin@172.16.100.2 \
-o StrictHostKeyChecking=accept-new \
sysadmin@192.168.100.12 \
"su -"

# Xocr altsrv3 (Nginx)
ssh -t \
-i ~/.ssh/id_alt-domain_2026_host_ed25519 \
-J sysadmin@172.16.100.2 \
-o StrictHostKeyChecking=accept-new \
sysadmin@192.168.100.14 \
"su -"

# Xocr altsrv4 (Samba-server1)
ssh -t \
-i ~/.ssh/id_alt-domain_2026_host_ed25519 \
-J sysadmin@172.16.100.2 \
-o StrictHostKeyChecking=accept-new \
sysadmin@192.168.100.14 \
"su -"

# Xocr altsrv5 (Samba-server2)
ssh -t \
-i ~/.ssh/id_alt-domain_2026_host_ed25519 \
-J sysadmin@172.16.100.2 \
-o StrictHostKeyChecking=accept-new \
sysadmin@192.168.100.15 \
"su -"

# Xocr altwks2
ssh -t \
-i ~/.ssh/id_alt-domain_2026_host_ed25519 \
-J sysadmin@172.16.100.2 \
-o StrictHostKeyChecking=accept-new \
sysadmin@192.168.100.2 \
"su -"
```

Подготовка для работы

```
# Регистрация сгенерированного ssh агентом
eval $(ssh-agent) \
&& ssh-add \
~/.ssh/id_alt-domain_2026_host_ed25519

# Вход на Хост altwks1
> ~/.ssh/known_hosts \
```

```
&& ssh -t -o StrictHostKeyChecking=accept-new \  
sysadmin@172.16.100.2  
  
# Проверяем наличие пары ключей ssh на altwks1  
find /home/sysadmin/.ssh/ \  
| grep alt-domain
```

- Проверка наличия пары ssh

Выполнение работы

Вход на сервер dc2 что является владельцем FSMO

```
ssh -t \  
-i ~/.ssh/id_alt-domain_2026_host_ed25519 \  
-J sysadmin@172.16.100.2 \  
-o StrictHostKeyChecking=accept-new \  
sysadmin@192.168.100.12 \  
"su -"
```

Проверка что сервер является владельцем fsmo

```
samba-tool fsmo show \  
| cut -d ',' -f 1-2
```

- Лог проверки что сервер является владельцем fsmo

Обновление пакетов и Установка пакетов admx-политик

```
apt-get update \  
&& apt-get -y install \  
admx-* \  
tree \  
&& admx-msi-setup
```

Перенос установленных политик в `/var/lib/samba/sysvol`

```
kinit -V Administrator \  
&& samba-tool gpo \  
admxload \  
--use-krb5-ccache=/tmp/krb5cc_0
```

- Лог переноса установленных политик в ``var/lib/samba/sysvol``

отображение списка политик командой find

```
find /var/lib/samba/sysvol | tail
```

► Лог отображения списка политик командой find

Проверка синхронизации sysvol ранее созданной службой dc1

```
journalctl -eu sysvol-syn.service
```

► Лог проверки синхронизации sysvol ранее созданной службой на dc1

```
[root@dc2 ~]# find /var/lib/samba/sysvol | tail
/var/lib/samba/sysvol/den.skv/Policies/PolicyDefinitions/nb-no/DataCollection.adml
/var/lib/samba/sysvol/den.skv/Policies/PolicyDefinitions/nb-no/Msi-FileRecovery.adml
/var/lib/samba/sysvol/den.skv/Policies/PolicyDefinitions/nb-no/CredentialProviders.adml
/var/lib/samba/sysvol/den.skv/Policies/PolicyDefinitions/nb-no/ExploitGuard.adml
/var/lib/samba/sysvol/den.skv/Policies/PolicyDefinitions/nb-no/WindowsBackup.adml
/var/lib/samba/sysvol/den.skv/Policies/PolicyDefinitions/nb-no/Display.adml
/var/lib/samba/sysvol/den.skv/Policies/PolicyDefinitions/nb-no/MobilePCMobilityCenter.adml
/var/lib/samba/sysvol/den.skv/Policies/PolicyDefinitions/nb-no/WindowsRemoteShell.adml
/var/lib/samba/sysvol/den.skv/Policies/PolicyDefinitions/nb-no/DnsClient.adml
/var/lib/samba/sysvol/den.skv/Policies/PolicyDefinitions/DataCollection.admx
[root@dc2 ~]# tree /var/lib/samba/sysvol | tail -1
38 directories, 5877 files
[root@dc2 ~]# cat /var/log/sysvol-sync.log | grep created | tail
[root@dc2 ~]#

[root@dc1 ~]# find /var/lib/samba/sysvol | tail
/var/lib/samba/sysvol/den.skv/Policies/PolicyDefinitions/nb-no/DataCollection.adml
/var/lib/samba/sysvol/den.skv/Policies/PolicyDefinitions/nb-no/Msi-FileRecovery.adml
/var/lib/samba/sysvol/den.skv/Policies/PolicyDefinitions/nb-no/CredentialProviders.adml
/var/lib/samba/sysvol/den.skv/Policies/PolicyDefinitions/nb-no/ExploitGuard.adml
/var/lib/samba/sysvol/den.skv/Policies/PolicyDefinitions/nb-no/WindowsBackup.adml
/var/lib/samba/sysvol/den.skv/Policies/PolicyDefinitions/nb-no/Display.adml
/var/lib/samba/sysvol/den.skv/Policies/PolicyDefinitions/nb-no/MobilePCMobilityCenter.adml
/var/lib/samba/sysvol/den.skv/Policies/PolicyDefinitions/nb-no/WindowsRemoteShell.adml
/var/lib/samba/sysvol/den.skv/Policies/PolicyDefinitions/nb-no/DnsClient.adml
/var/lib/samba/sysvol/den.skv/Policies/PolicyDefinitions/DataCollection.admx
[root@dc1 ~]# tree /var/lib/samba/sysvol | tail -1
38 directories, 5877 files
[root@dc1 ~]# cat /var/log/sysvol-sync.log | grep created | tail
The file sysvol/den.skv/Policies/PolicyDefinitions/tr-tr/sdiagschd.adml has been created
The file sysvol/den.skv/Policies/PolicyDefinitions/tr-tr/srm-fci.adml has been created
The file sysvol/den.skv/Policies/PolicyDefinitions/tr-tr/tcpi.adml has been created
The file sysvol/den.skv/Policies/PolicyDefinitions/tr-tr/terminalserver-server.adml has been created
The file sysvol/den.skv/Policies/PolicyDefinitions/tr-tr/windowserver.adml has been created
The file sysvol/den.skv/Policies/PolicyDefinitions/tr-tr/wlansvc.adml has been created
The file sysvol/den.skv/Policies/PolicyDefinitions/zh-TW/thunderbird.adml has been created
The file sysvol/den.skv/Policies/PolicyDefinitions/zh-TW/wlansvc.adml has been created
The file sysvol/den.skv/Policies/PolicyDefinitions/zh-TW/wmansvc.adml has been created
[root@dc1 ~]# systemctl cat sysvol-syn.service
# /etc/systemd/system/sysvol-syn.service
[Unit]
Description=Sysvol sync
After=network.target

[Service]
Type=oneshot
Environment=HOME=/root
Environment=SSH_AUTH_SOCK=/run/sysvol-ssh-agent.sock
ExecStartPre=/usr/bin/ssh-agent -a /run/sysvol-ssh-agent.sock
ExecStart=/usr/bin/ssh-add /root/.ssh/id_sysvol_ed25519
ExecStart=/usr/bin/unison sync_dc2 -silent
ExecStartPost=/usr/bin/pkill -f "ssh-agent -a /run/sysvol-ssh-agent.sock"
```

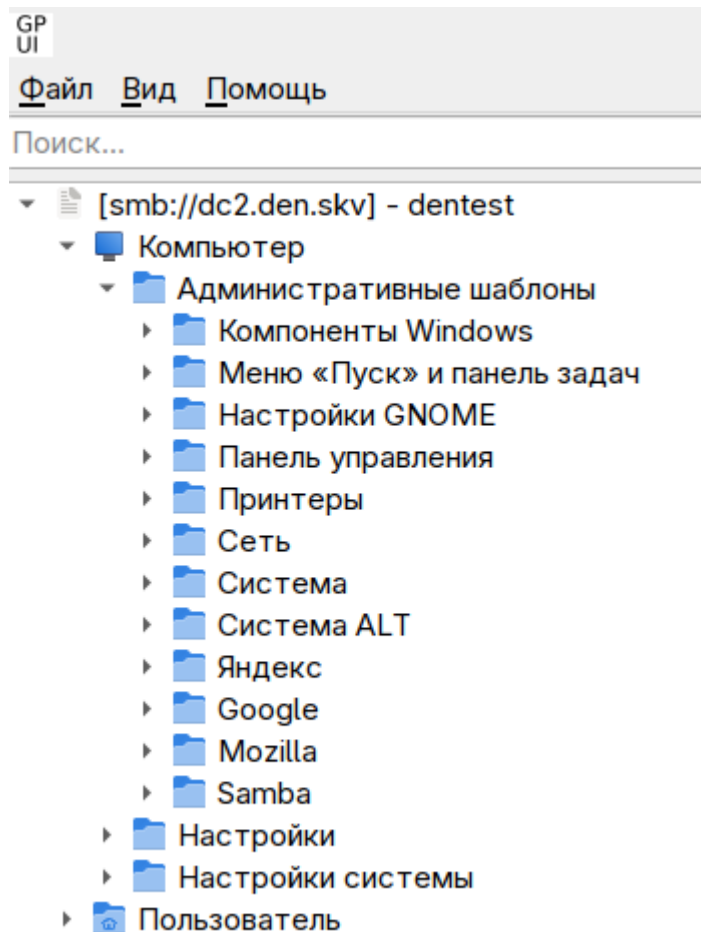
Вход на рабочую станцию введенную в домен altwks2

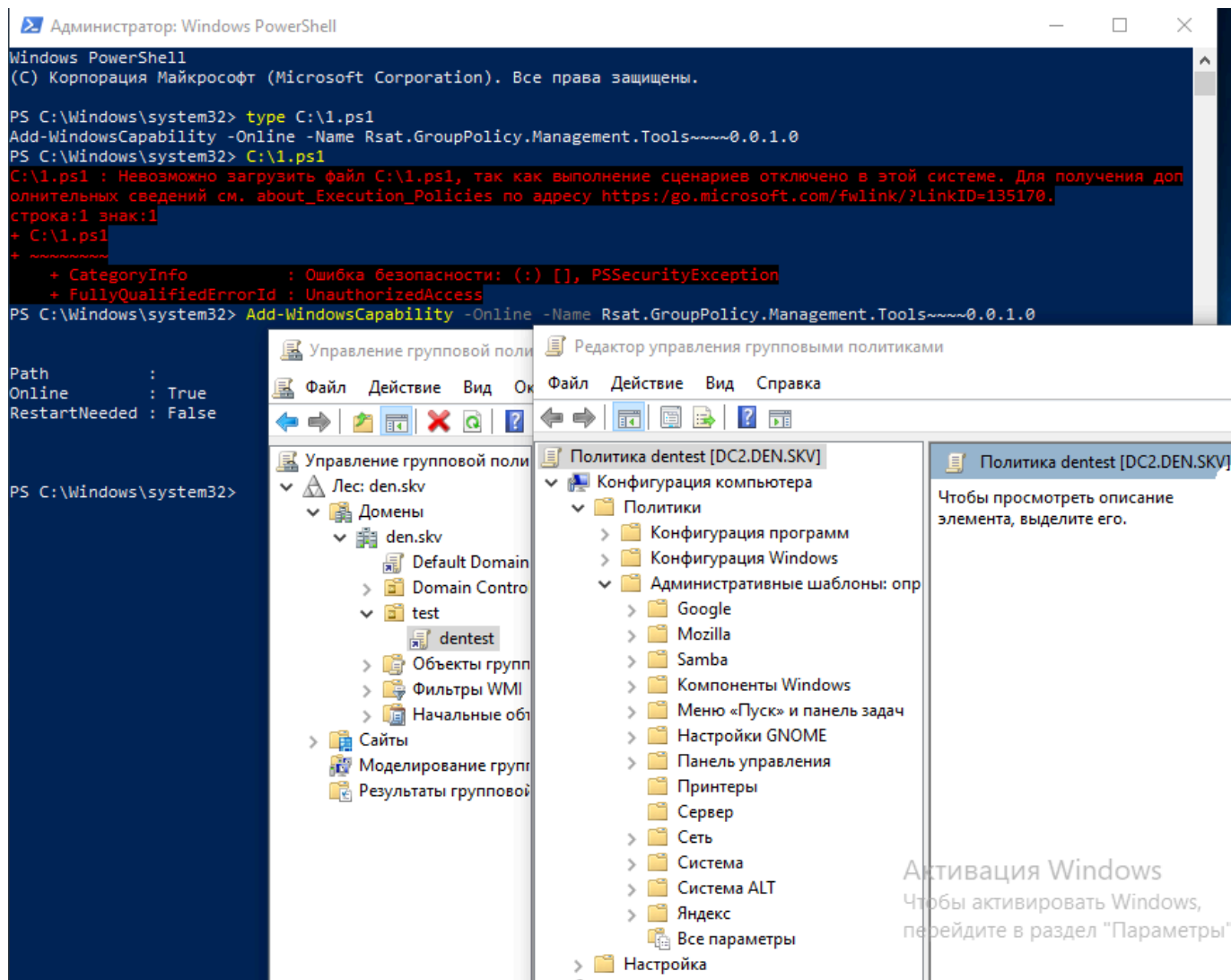
```
ssh -t \
-i ~/.ssh/id_alt-domain_2026_host_ed25519 \
-J sysadmin@172.16.100.2 \
-o StrictHostKeyChecking=accept-new \
sysadmin@192.168.100.2 \
"su -"
```

Обновление пакетов и Установка пакетов управления\клиентских\диагностических утилит групповых политик

```
apt-get update \
&& apt-get -y install \
admx-* \
tree \
admc \
```

```
gpui \  
gpupdate \  
alterator-gpupdate \  
gpresult \  
&& admx-msi-setup
```





Включение применения групповых политик и ввод в обозначенный режим workstation

Вывод доступного списка профилей групповых политик

```
gpupdate-setup list
```

► Вывод доступного списка профилей групповых политик

Вывод текущего режима групповых политик

```
gpupdate-setup status \
&& gpupdate-setup active-policy \
&& gpupdate-setup default-policy
```

► Вывод текущего режима групповых политик

Применение и активация профиля групповых политик в workstation

```
gpupdate-setup \  
write \  
enable \  
workstation \  
&& systemctl daemon-reload
```

- Применение и активация профиля групповых политик в `workstation`

Вывод режима групповых политик

```
gpupdate-setup status \  
&& gpupdate-setup active-policy \  
&& gpupdate-setup default-policy
```

- Вывод текущего режима групповых политик

```
[root@altwks2 ~]# gpupdate-setup list  
workstation  
ad-domain-controller  
server  
[root@altwks2 ~]# gpupdate-setup status \  
&& gpupdate-setup active-policy \  
&& gpupdate-setup default-policy  
disabled  
default  
workstation  
[root@altwks2 ~]# gpupdate-setup \  
write \  
enable \  
workstation  
workstation  
Warning: The unit file, source configuration file or drop-ins of autofs.service changed on disk. Ru  
n 'systemctl daemon-reload' to reload units.  
Created symlink '/etc/systemd/system/multi-user.target.wants/gpupdate-scripts-run.service' → '/usr/  
lib/systemd/system/gpupdate-scripts-run.service'.  
Created symlink '/etc/systemd/user/default.target.wants/gpupdate-scripts-run-user.service' → '/usr/  
lib/systemd/user/gpupdate-scripts-run-user.service'.  
Created symlink '/etc/systemd/system/timers.target.wants/gpupdate.timer' → '/usr/lib/systemd/system  
/gpupdate.timer'.  
Created symlink '/etc/systemd/user/timers.target.wants/gpupdate-user.timer' → '/usr/lib/systemd/use  
r/gpupdate-user.timer'.  
[root@altwks2 ~]# systemctl daemon-reload  
[root@altwks2 ~]# gpupdate-setup status \  
&& gpupdate-setup active-policy \  
&& gpupdate-setup default-policy  
enabled  
workstation  
workstation  
[root@altwks2 ~]#
```

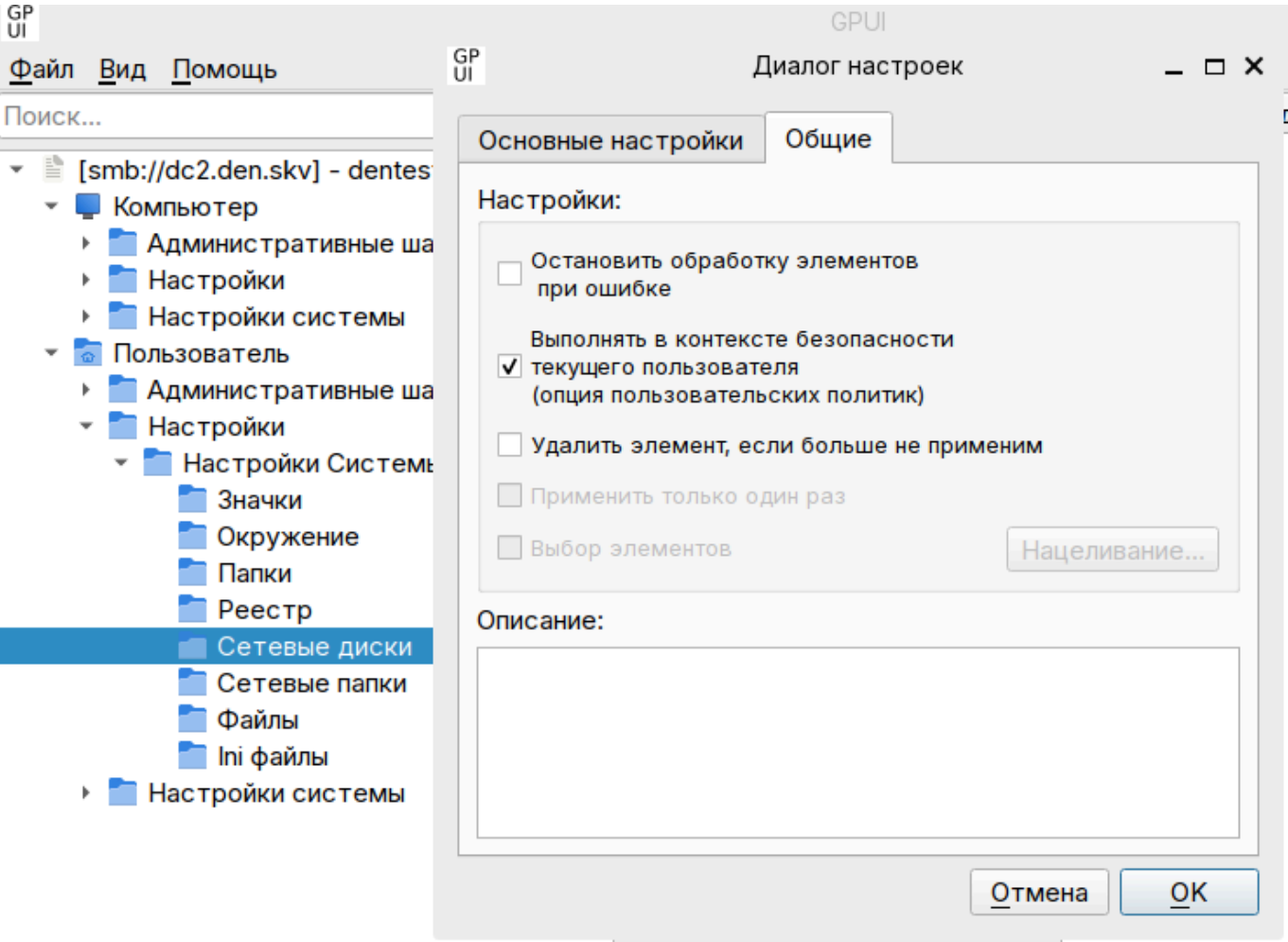
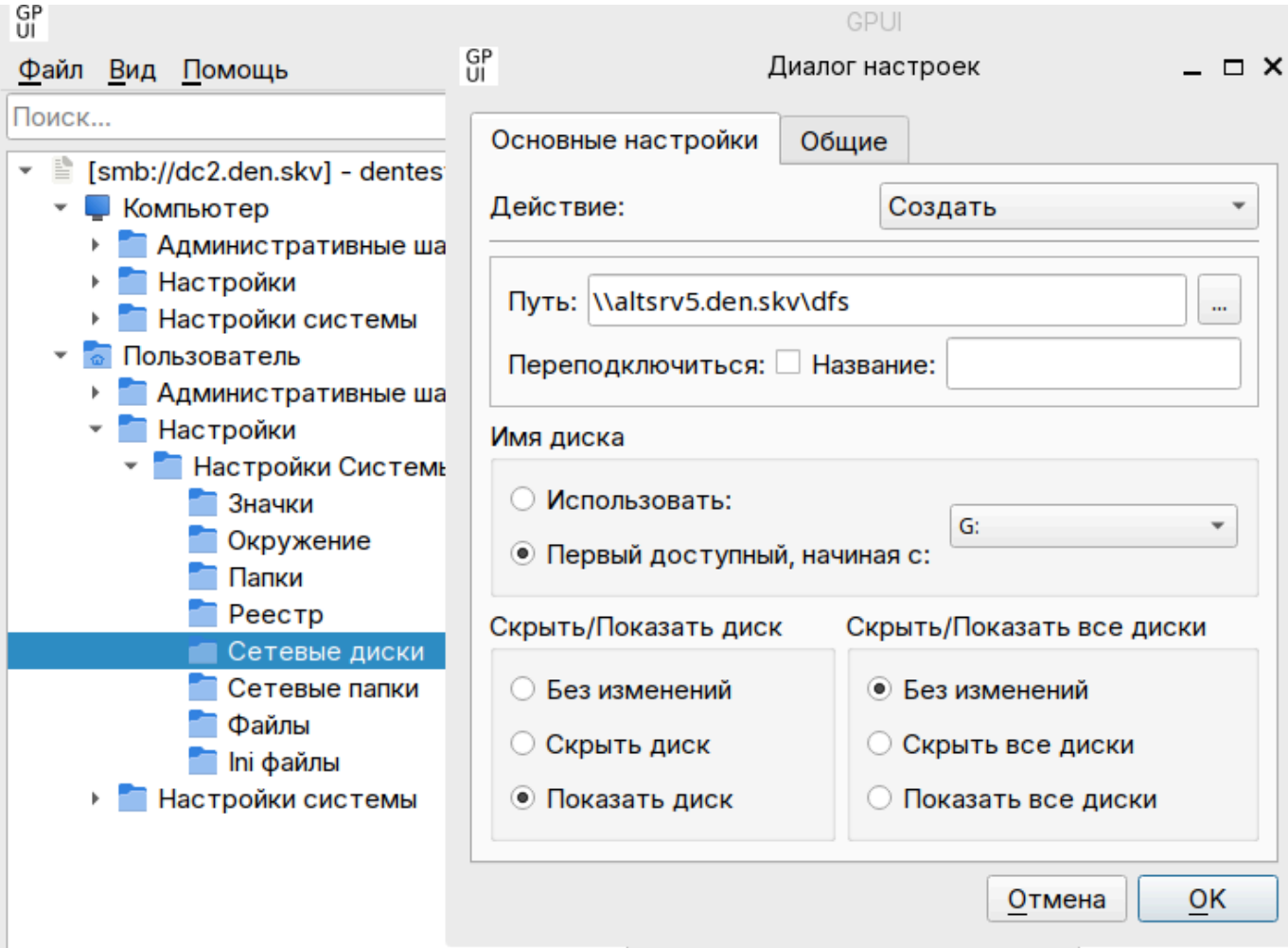
Копирование файлов по ssh через jump-proxy

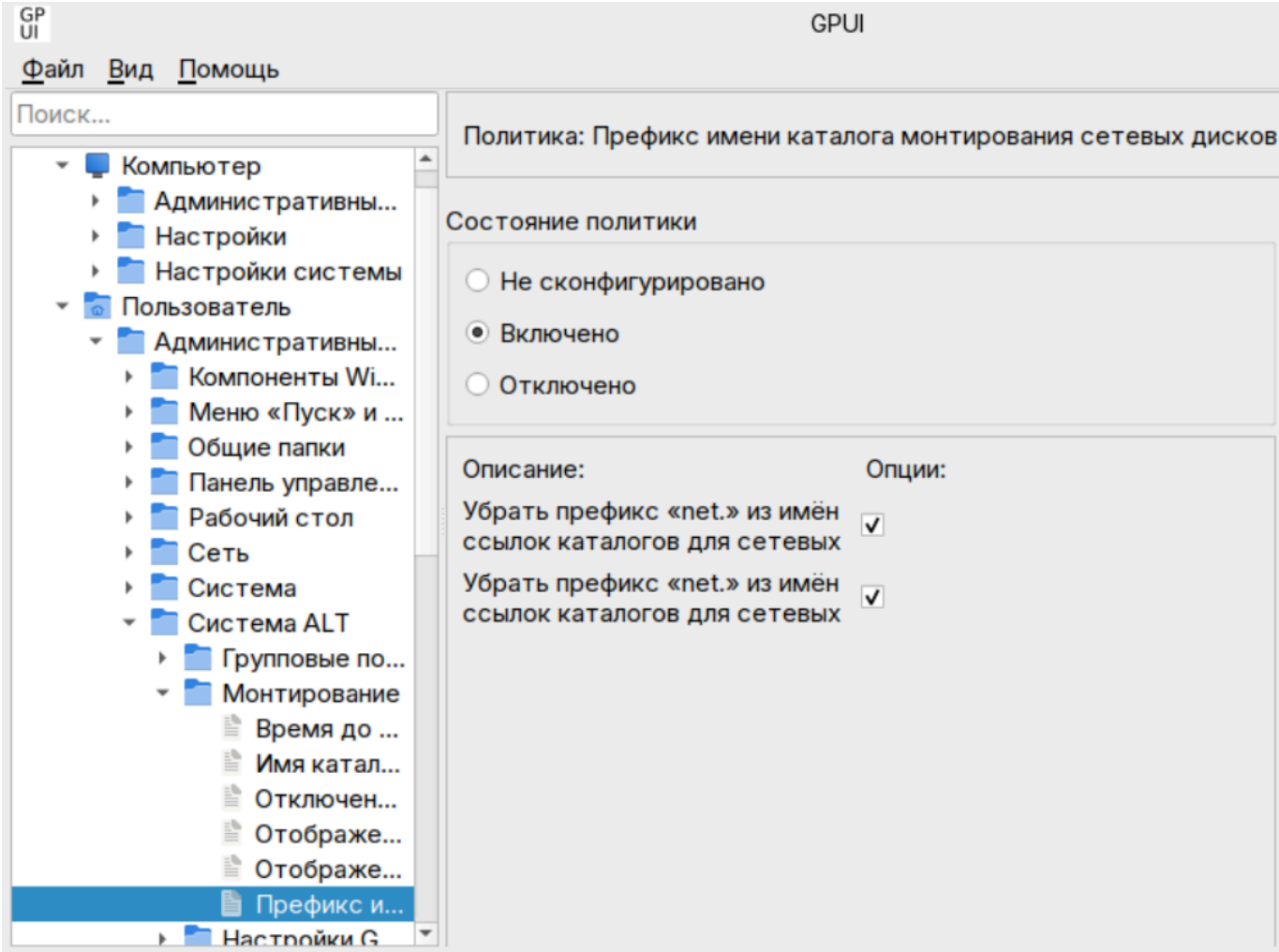
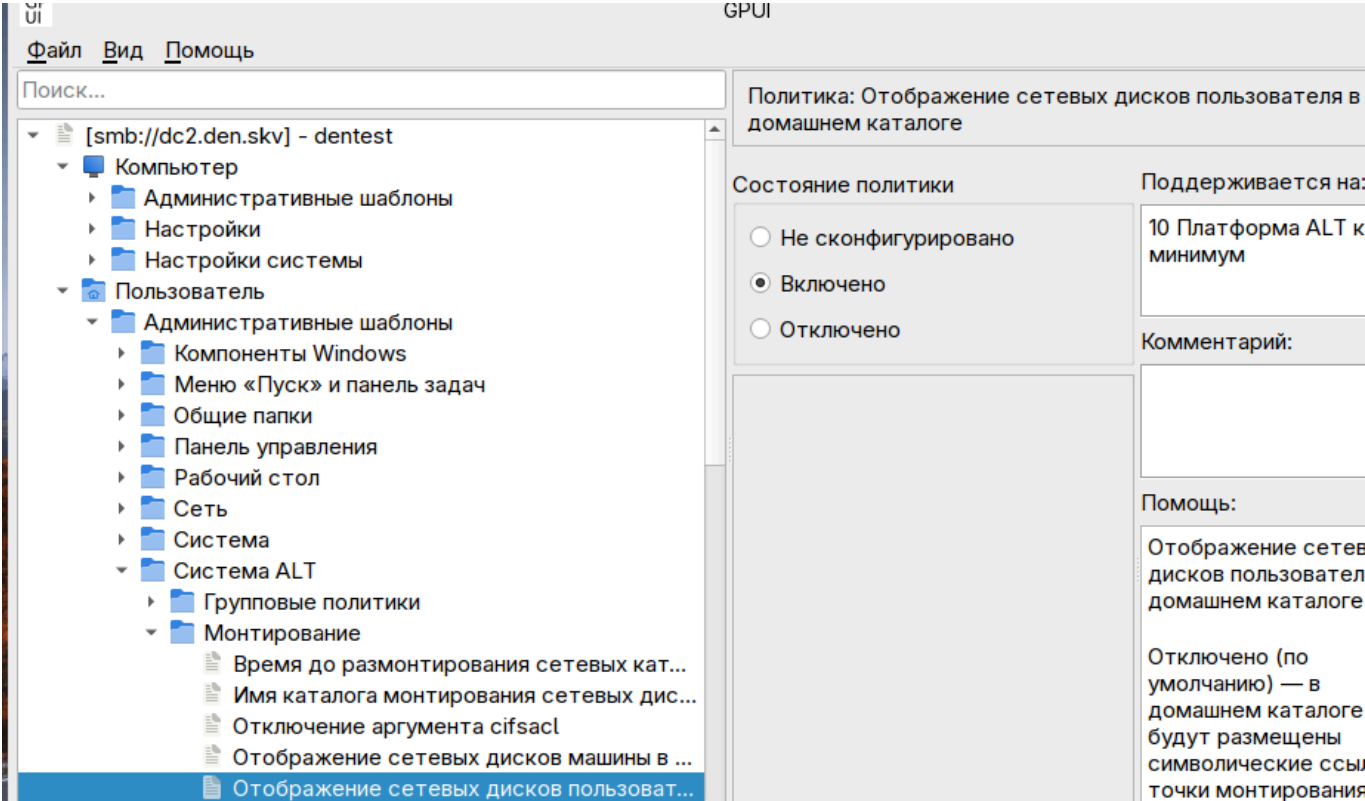
```
scp -i ~/.ssh/id_alt-domain_2026_host_ed25519 \  
-o StrictHostKeyChecking=accept-new \  
-J sysadmin@172.16.100.2 \  
-
```

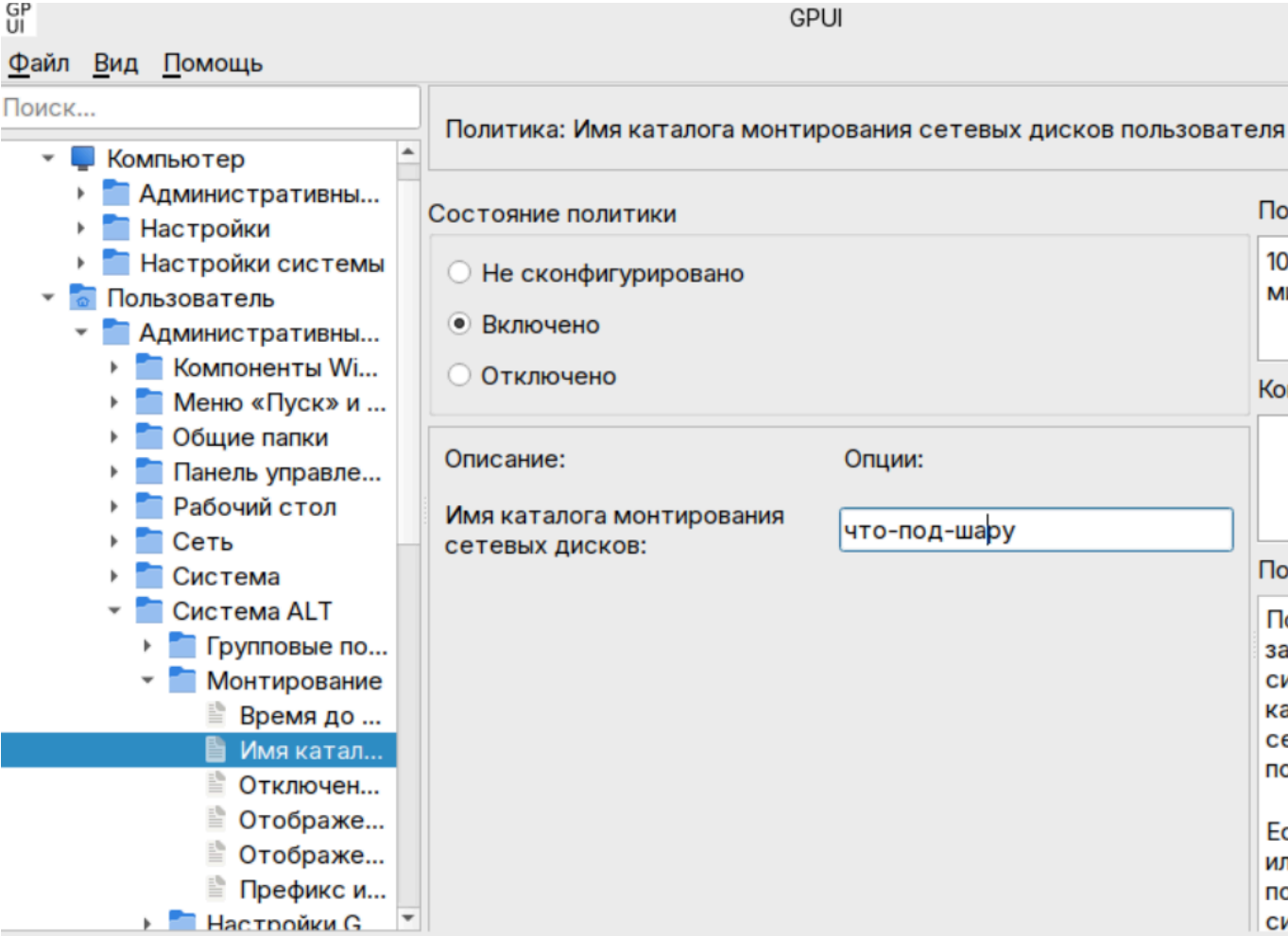


```
./WindowsTH-KB2693643-x64.msu \  
samba_u1@192.168.100.14:/srv/samba/dfs/WindowsTH-KB2693643-x64.msu
```

Создание политики подключения SMB-ресурса к сессии доменного пользователя

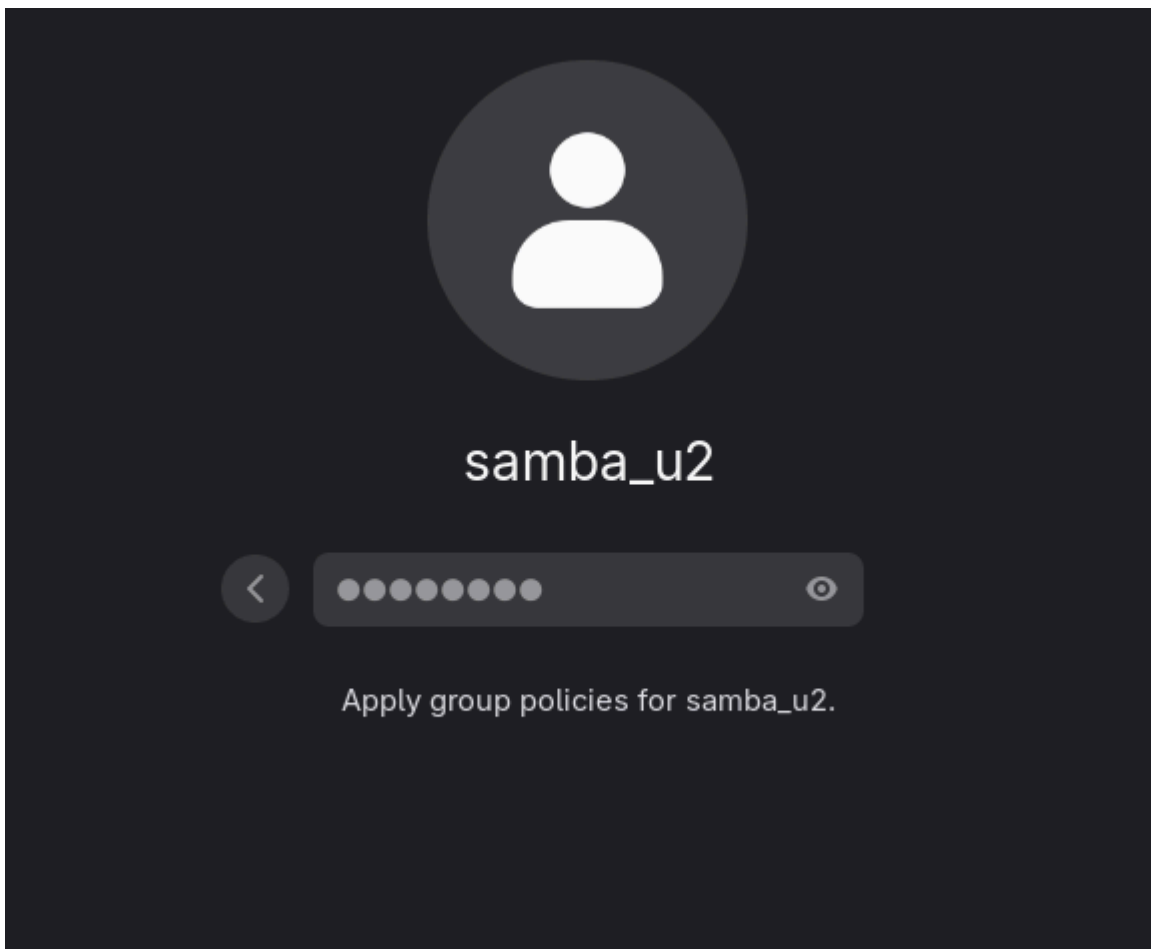
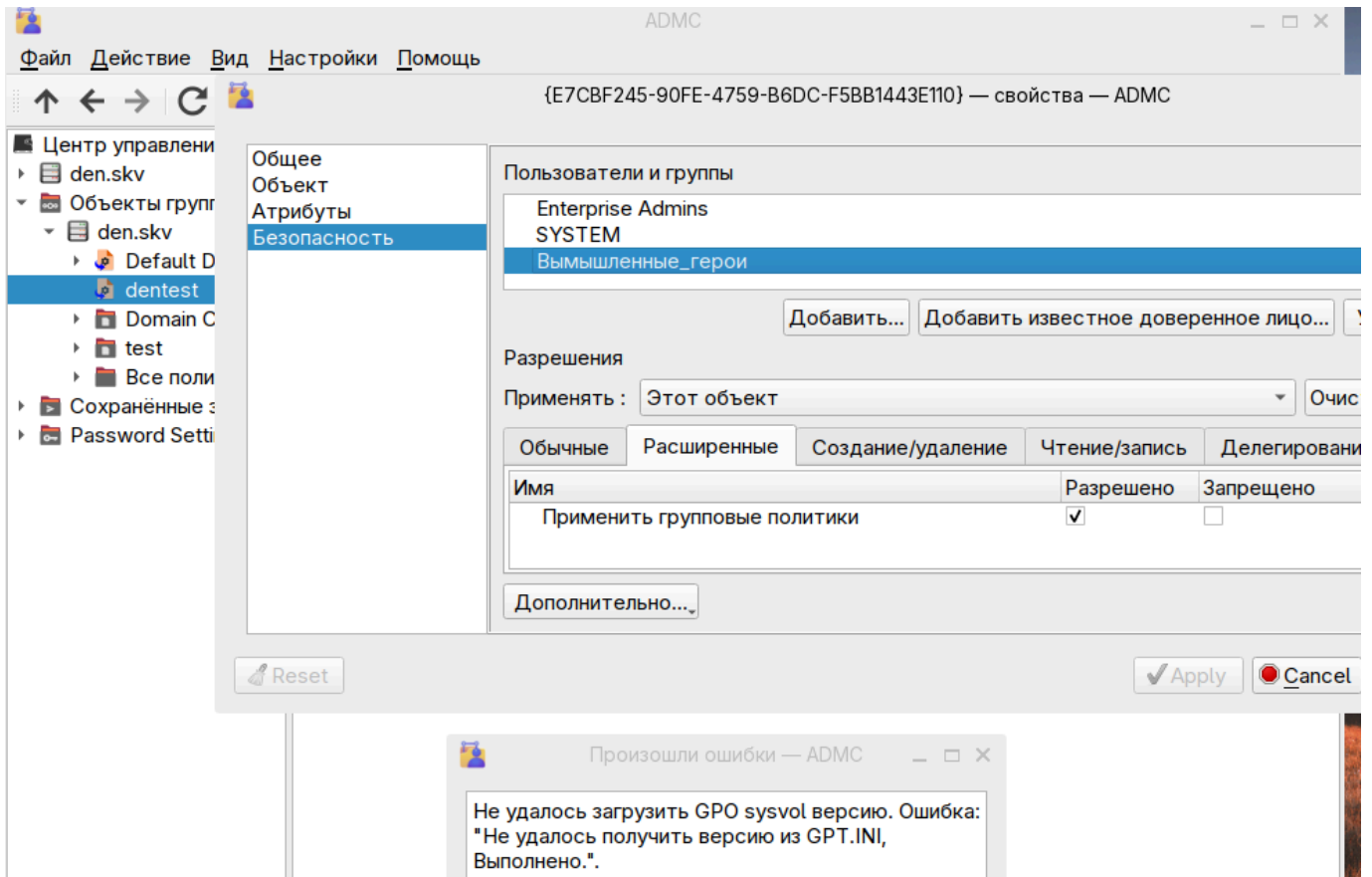


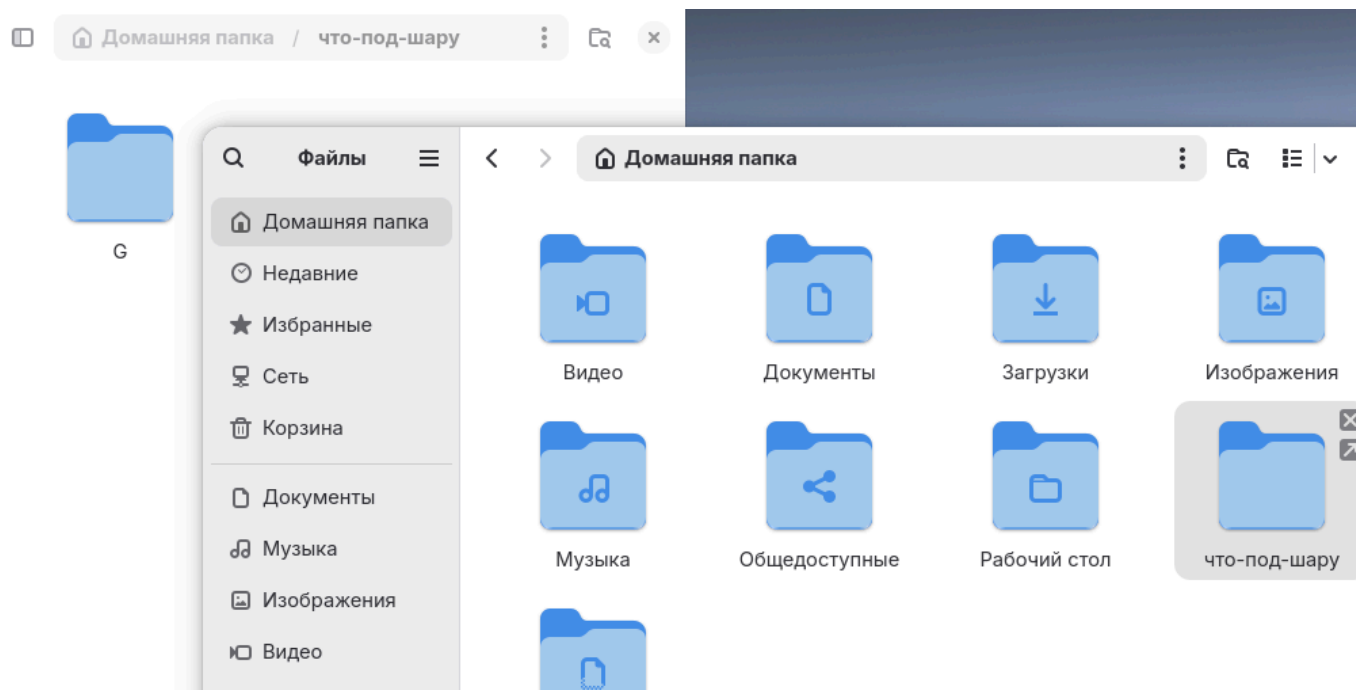




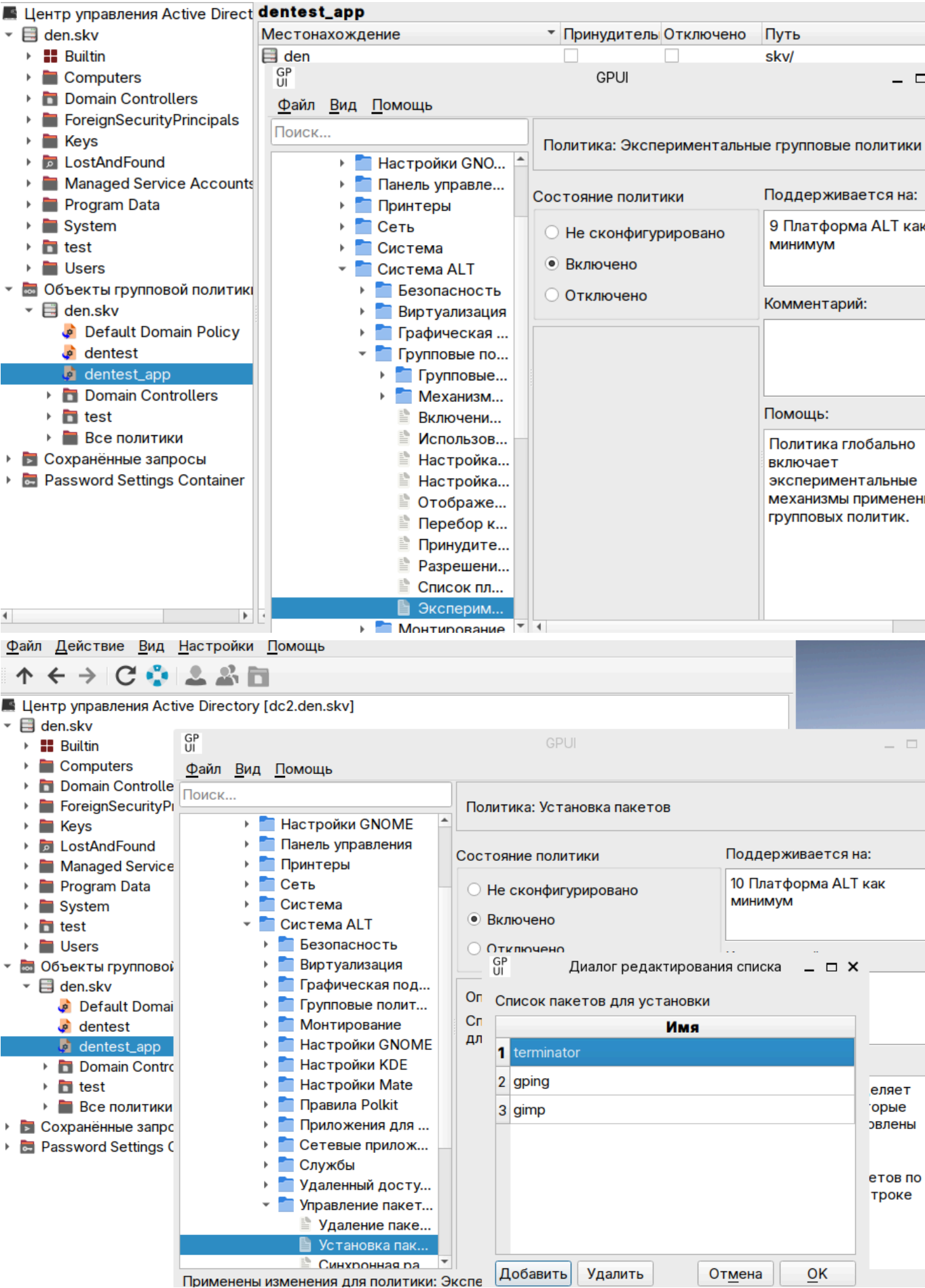
Применение политики для группы

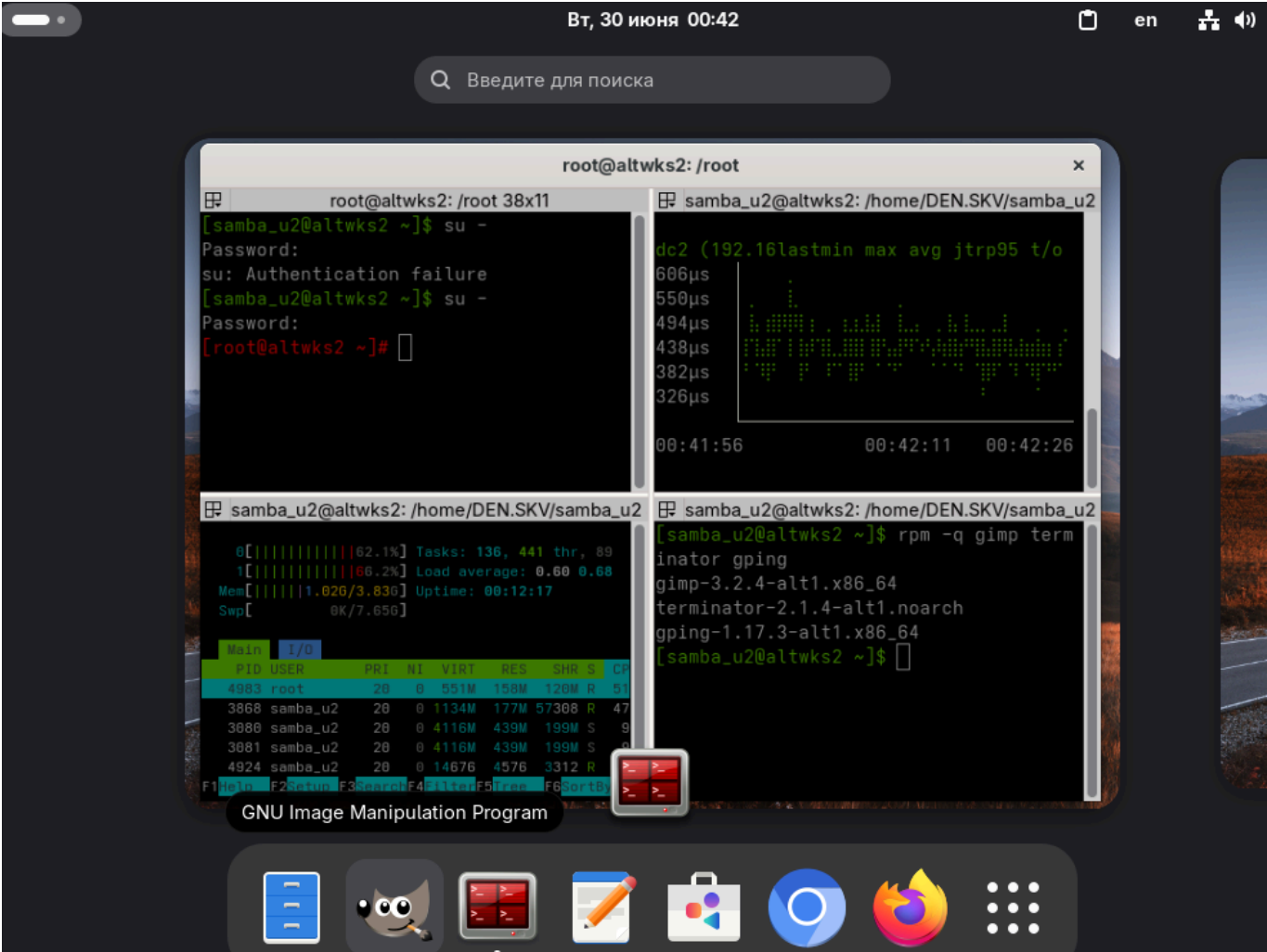
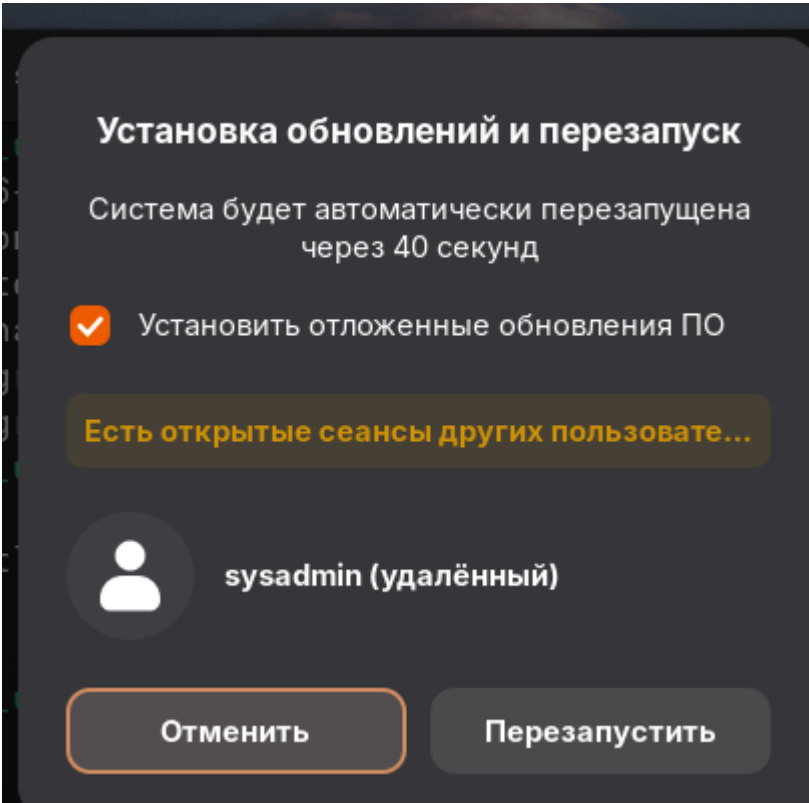
Сетевые папки





Установка пакетов





Пересоздание файла `/etc/sudoers`

Вход на Домен контролер с FSMO


```
ssh -t \  
-i ~/.ssh/id_alt-domain_2026_host_ed25519 \  
-J sysadmin@172.16.100.2 \  
-o StrictHostKeyChecking=accept-new \  
sysadmin@192.168.100.12 \  
"su -"
```

Создать каталог для копирования и скопировать отредактированный `/etc/sudoers`

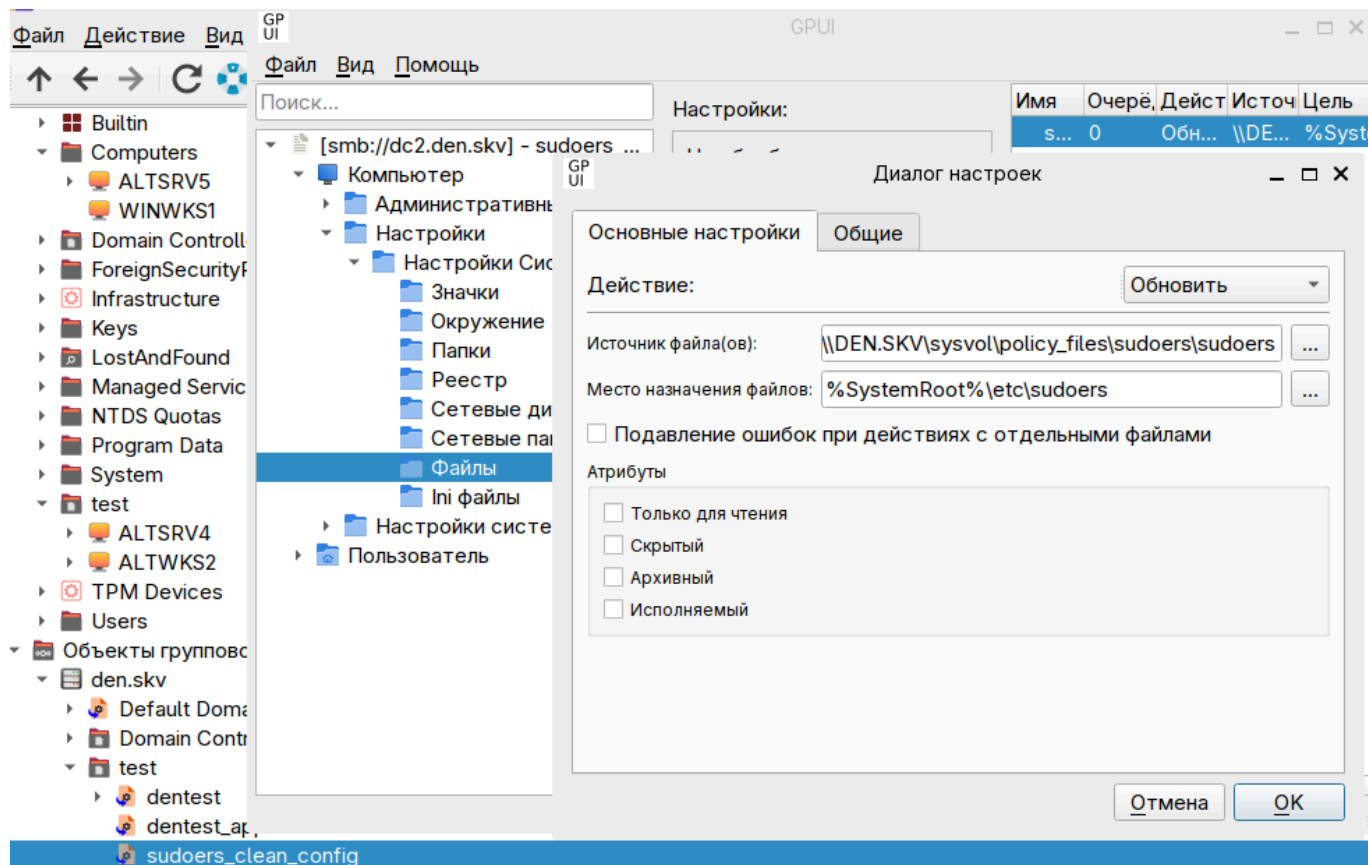
```
mkdir -vp \  
/var/lib/samba/sysvol/policy_files/sudoers  
  
cp -v /etc/sudoers \  
/var/lib/samba/sysvol/policy_files/sudoers/  
  
sed -i \  
-e '/^[[[:space:]]*#/d' \  
-e '/^[[[:space:]]*$/d' \  
/var/lib/samba/sysvol/policy_files/sudoers/sudoers  
  
cat /var/lib/samba/sysvol/policy_files/sudoers/sudoers  
  
chmod -v o+r -R /var/lib/samba/sysvol/policy_files
```

► Вывод текущего содержимого файла `/etc/sudoers`

Центр управления Active Directory [dc2.den.skv]

- ▼ den.skv
 - ▶ Built-in
 - ▼ Computers
 - ▶ ALTSRV5
 - ▶ WINWKS1
 - ▶ Domain Controllers
 - ▶ ForeignSecurityPrincipals
 - ▶ Infrastructure
 - ▶ Keys
 - ▶ LostAndFound
 - ▶ Managed Service Accounts
 - ▶ NTDS Quotas
 - ▶ Program Data
 - ▶ System
 - ▼ test
 - ▶ ALTSRV4
 - ▶ ALTWKS2
 - ▶ TPM Devices
 - ▶ Users
 - ▼ Объекты групповой политики
 - ▼ den.skv
 - ▶ Default Domain Policy
 - ▶ Domain Controllers
 - ▼ test
 - ▶ dentest
 - ▶ dentest_app
 - ▶ sudoers_clean_config

```
[root@dc2 ~]# mkdir -vp \
/var/lib/samba/sysvol/policy_files/sudoers
mkdir: создан каталог '/var/lib/samba/sysvol/policy_files'
mkdir: создан каталог '/var/lib/samba/sysvol/policy_files/sudoers'
[root@dc2 ~]# cp -v /etc/sudoers \
/var/lib/samba/sysvol/policy_files/sudoers/
'/etc/sudoers' -> '/var/lib/samba/sysvol/policy_files/sudoers/sudoers'
[root@dc2 ~]# sed -i \
-e '/^[[[:space:]]*#/d' \
-e '/^[[[:space:]]*$/d' \
/var/lib/samba/sysvol/policy_files/sudoers/sudoers
[root@dc2 ~]# cat /var/lib/samba/sysvol/policy_files/sudoers/sudoers
User_Alias    WHEEL_USERS = %wheel
User_Alias    XGRP_USERS = %xgrp
Defaults!/usr/sbin/visudo env_keep += "SUDO_EDITOR EDITOR VISUAL"
Defaults secure_path="/sbin:/usr/sbin:/usr/local/sbin:/bin:/usr/bin:/usr/local/bin"
Defaults:XGRP_USERS env_keep += "DISPLAY XAUTHORITY"
@includedir /etc/sudoers.d
[root@dc2 ~]# chmod -v o+r -R /var/lib/samba/sysvol/policy_files
права доступа '/var/lib/samba/sysvol/policy_files' оставлены в виде 0755 (rwxr-xr-x)
права доступа '/var/lib/samba/sysvol/policy_files/sudoers' оставлены в виде 0755 (rwxr-xr-x)
права доступа '/var/lib/samba/sysvol/policy_files/sudoers/sudoers' изменены с 0400 (r-----) на 0404 (r-----r--)
[root@dc2 ~]#
```



```
[root@altsrv4 ~]# gpupdate-setup status \
&& gpupdate-setup active-policy
enabled
server
[root@altsrv4 ~]# tail -5 /etc/sudoers
# Defaults targetpw # Ask for the password of the target user
# ALL ALL=(ALL:ALL) ALL # WARNING: only use this together with 'Defaults t
argetpw'

## Read drop-in files from /etc/sudoers.d
@includedir /etc/sudoers.d
[root@altsrv4 ~]# gpoa --dc dc2 --force
[root@altsrv4 ~]# tail -5 /etc/sudoers
User_Alias XGRP_USERS = %xgrp
Defaults!usr/sbin/visudo env_keep += "SUDO_EDITOR EDITOR VISUAL"
Defaults secure_path="/sbin:/usr/sbin:/usr/local/sbin:/bin:/usr/bin:/usr/lo
cal/bin"
Defaults:XGRP_USERS env_keep += "DISPLAY XAUTHORITY"
@includedir /etc/sudoers.d
[root@altsrv4 ~]# ll !$
ll /etc/sudoers
-rw-r--r-- 1 root root 292 июн 30 23:26 /etc/sudoers
[root@altsrv4 ~]# date
Вт 30 июн 2026 23:27:29 MSK
[root@altsrv4 ~]#
```

```
[root@altwks2 ~]# gpupdate-setup status \
&& gpupdate-setup active-policy
enabled
workstation
[root@altwks2 ~]# tail -5 /etc/sudoers
# Defaults targetpw # Ask for the password of the target user
# ALL ALL=(ALL:ALL) ALL # WARNING: only use this together with 'Default
s targetpw'

## Read drop-in files from /etc/sudoers.d
@includedir /etc/sudoers.d
[root@altwks2 ~]# gpoa --dc dc2 --force
[root@altwks2 ~]# tail -5 /etc/sudoers
User_Alias XGRP_USERS = %xgrp
Defaults!usr/sbin/visudo env_keep += "SUDO_EDITOR EDITOR VISUAL"
Defaults secure_path="/sbin:/usr/sbin:/usr/local/sbin:/bin:/usr/bin:/usr
/local/bin"
Defaults:XGRP_USERS env_keep += "DISPLAY XAUTHORITY"
@includedir /etc/sudoers.d
[root@altwks2 ~]# ll !$
ll /etc/sudoers
-rw-r--r-- 1 root root 292 июн 30 23:26 /etc/sudoers
[root@altwks2 ~]# date
Вт 30 июн 2026 23:27:25 MSK
[root@altwks2 ~]#
```

Обновление пакетов и Установка пакетов клиентских\диагностических утилит групповых политик для altsrv4

```
apt-get update \
&& apt-get -y install \
tree \
gpupdate \
gpreresult \
xdg-user-dirs
```

Применение и активация профиля групповых политик в server

```
gpupdate-setup \  
write \  
enable \  
server \  
&& systemctl daemon-reload
```

- Вывод текущего содержимого файла `/etc/sudoers`

Вывод режима групповых политик

```
gpupdate-setup status \  
&& gpupdate-setup active-policy
```

- Вывод текущего режима групповых политик

Обновление таймера обновления групповых политик

до применения политик

```
systemctl cat gpupdate.timer
```

- Вывод текущего содержимого файла `/etc/systemd/system/gpupdate.timer`

↑ ← → ↻

Файл Вид Помощь

Поиск...

Бuiltin

Computers

ALTSRV5

WINWKS1

Domain Cont

ForeignSecur

Infrastructure

Keys

LostAndFour

Managed Ser

NTDS Quotas

Program Dat

System

test

ALTSRV4

ALTWKS2

TPM Device

Users

Объекты групп

den.skv

Default D

Domain C

test

dentes

dentes

gpo_applier_timer_30min

sudoers_clean_config

Все политики

Сохранённые запросы

[smb://dc2.den.skv] - gpo_applier...

Компьютер

Административные шаблоны

Настройки

Настройки Системы

Значки

Окружение

Папки

Реестр

Сетевые диски

Сетевые папки

Файлы

Ini файлы

Настройки системы

Пользователь

Настройки:

Не обра... GP UI

эле... GP UI

Залу... GP UI

конт... GP UI

поль... GP UI

Удал... GP UI

не п... GP UI

При... GP UI

оди... GP UI

Отф... GP UI

Отф... GP UI

урое... GP UI

Откл... GP UI

Откл... GP UI

упрс... GP UI

Описа... GP UI

Без с... GP UI

Имя	Очерё	Дейст	Путь	Секци	Свойс	Значен
g...	0	Обн...	/usr/...	Timer	OnS...	30min

Диалог настроек

Основные настройки

Общие

Действие:

Обновить

Путь к файлу

/usr/lib/systemd/system/gpupdate.timer

...

Имя секции

Timer

Имя свойства

OnStartupSec

Значение свойства

30min

Отмена

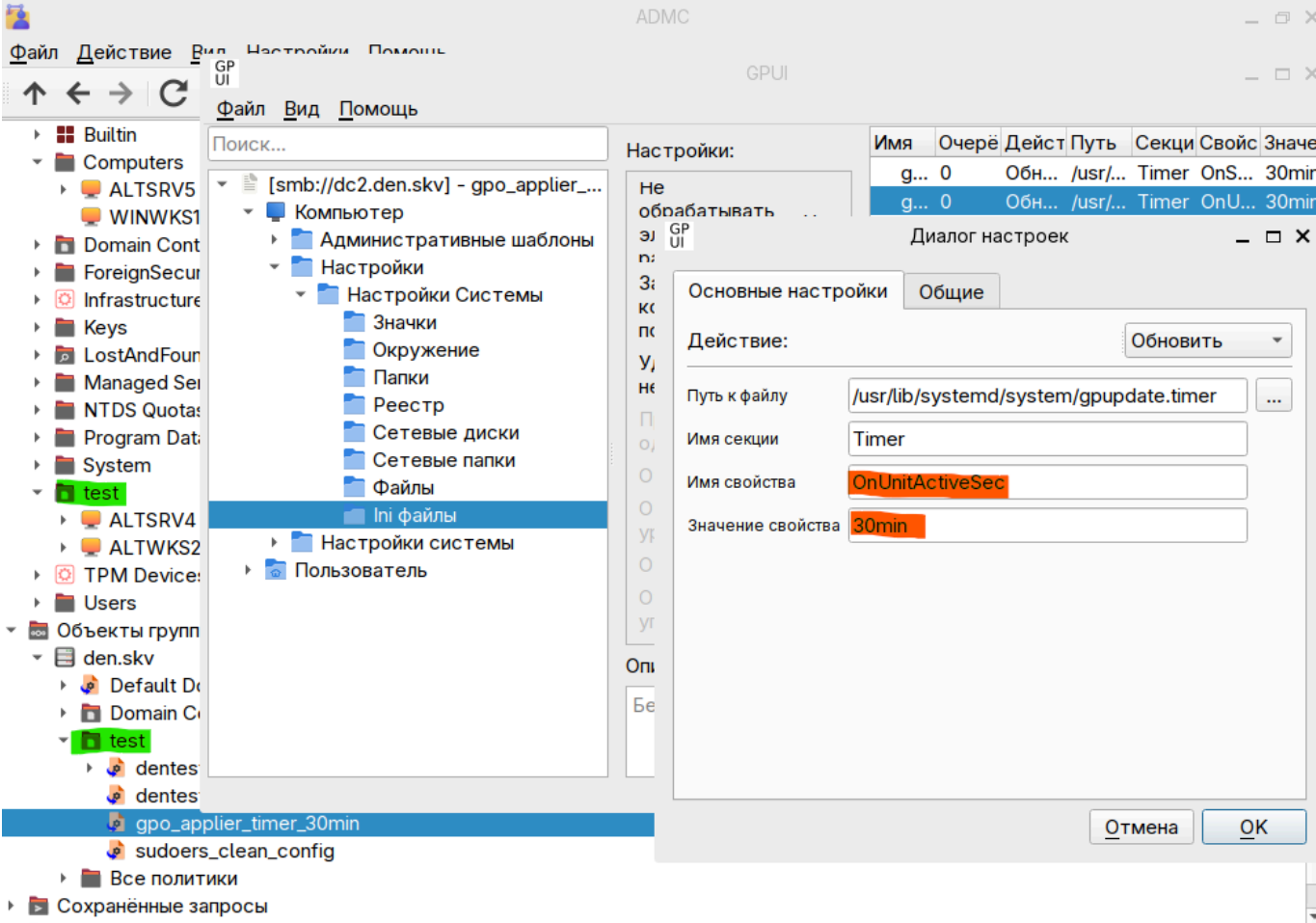
OK

administrator@den.skv

```
[root@altsrv4 ~]# systemctl cat gpupdate.timer
# /usr/lib/systemd/system/gpupdate.timer
[Unit]
Description=Run gpupdate every hour

[Timer]
OnStartupSec=60min
OnUnitActiveSec=60min

[Install]
WantedBy=timers.target
```



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[Unit]
Description=Run gpupdate every hour

[Timer]
OnStartupSec=60min
OnUnitActiveSec=60min

[Install]
WantedBy=timers.target
[root@altsrv4 ~]# gpoa --dc dc2 --force
[root@altsrv4 ~]# systemctl cat gpupdate.timer
# Warning: gpupdate.timer changed on disk, the version systemd has loaded
# This output shows the current version of the unit's original fragment ab
# If fragments or drop-ins were added or removed, they are not properly r
# Run 'systemctl daemon-reload' to reload units.
# /usr/lib/systemd/system/gpupdate.timer
[Unit]
Description = Run gpupdate every hour

[Timer]
OnStartupSec = 30min
OnUnitActiveSec = 30min

[Install]
WantedBy = timers.target
[root@altsrv4 ~]# systemctl daemon-reload
[root@altsrv4 ~]# systemctl cat gpupdate.timer
# /usr/lib/systemd/system/gpupdate.timer
[Unit]
Description = Run gpupdate every hour

[Timer]
OnStartupSec = 30min
OnUnitActiveSec = 30min

[Install]
WantedBy = timers.target
[root@altsrv4 ~]#
```

```
[root@altwks2 ~]# systemctl cat gpupdate.timer
# /usr/lib/systemd/system/gpupdate.timer
[Unit]
Description=Run gpupdate every hour

[Timer]
OnStartupSec=60min
OnUnitActiveSec=60min

[Install]
WantedBy=timers.target
[root@altwks2 ~]# gpoa --dc dc2 --force
[root@altwks2 ~]# systemctl cat gpupdate.timer
# Warning: gpupdate.timer changed on disk, the version systemd has loaded
# This output shows the current version of the unit's original fragment
# If fragments or drop-ins were added or removed, they are not properly
# Run 'systemctl daemon-reload' to reload units.
# /usr/lib/systemd/system/gpupdate.timer
[Unit]
Description = Run gpupdate every hour

[Timer]
OnStartupSec = 30min
OnUnitActiveSec = 30min

[Install]
WantedBy = timers.target
[root@altwks2 ~]# systemctl daemon-reload
[root@altwks2 ~]# systemctl cat gpupdate.timer
# /usr/lib/systemd/system/gpupdate.timer
[Unit]
Description = Run gpupdate every hour

[Timer]
OnStartupSec = 30min
OnUnitActiveSec = 30min

[Install]
WantedBy = timers.target
[root@altwks2 ~]#
```

после применения политик

```
systemctl cat gpupdate.timer
```

► Вывод текущего содержимого файла `/etc/systemd/system/gpupdate.timer`

Обновление таймера для systemd

```
systemctl daemon-reload
```

Для github и gitflirc

```
exit

git branch -v

git log --oneline

git switch main

git status

pushd \
..

git rm -r --cached \
. ../

git add . ../ \
&& git status

git remote -v

git commit -am "GPO" \
&& git push \
--set-upstream \
altlinux \
main \
&& git push \
--set-upstream \
altlinux_gf \
main \
&& git push \
--set-upstream \
altlinux_sc \
main

popd
```